MILITARY SPECIFICATION

ADHESIVE, COPOLYMER POLYURETHANE

This specification is approved for use within the Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of a two-part epoxy polyurethane copolymer adhesive.

1.2 Military part number. The military part number shall consist of the letter B, plus the basic number of this specification, followed by a combination of letters denoting container size (see table I).

Example: B 47318 PT

- PT — Container size (see table I)
- 47318 — Basic specification number
- B — Letter B denoting bulk material

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) at the end of this document or by letter.

AMSC N/A FSC 8040
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TABLE I. Container size.

Container size 1/	Container size designator
One ounce----------	OZ
One pint----------	PT
One quart----------	QT
One gallon----------	GL

1/ Wherever inch pound units of measurement are expressed in this specification, metric equivalents shall be acceptable.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. The following standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

MILITARY

MIL-STD-129 Marking for Shipment and Storage

MIL-STD-130 Identification Marking of U.S. Military Property

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D790	Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials, Standard Test Method for
ASTM D1002	Strength Properties of Adhesives in Shear by Tension Loading (Metal-to-Metal), Standard Test Method for
ASTM D1084	Viscosity of Adhesives, Standard Test Method for
ASTM D2240	Rubber Property - Durometer Hardness, Standard Test Method for
ASTM D3951	Commercial Packaging, Standard Practice for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. The documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article inspection. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.2, and 6.3).

3.2 Material. Material shall be an epoxy copolymer urethane adhesive base (part A) and a hardening agent (part B), in separate containers clearly indicating material (see 4.6.1).

3.3 Cure time. The material shall cure in not more than four hours at a temperature of 65 ± 5°Celsius(C) or 149 ± 9°Fahrenheit(F). See 4.6.2.

3.4 Pot life. Pot life at 25 ± 2°C (77 ± 4°F) shall be not greater than one hour (see 4.6.3).

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3.5 Viscosity. Immediately after mixing, the viscosity shall be 2000 to 4000 centipoises (see 4.6.4).

3.6 Hardness. Shore D hardness shall be not less than 70 at 25 ± 5°C (77 ± 9°F) (see 4.6.5).

3.7 Shear strength. Lap shear strength shall be not less than 2600 pounds per square inch (lbf/in^2) (see 4.6.6).

3.8 Flexural strength. Flexural strength shall be not less than 8500 lbf/in^2 (see 4.6.7).

3.9 Shelf life. When stored in unopened containers at a temperature not greater than 25°C (77°F) for a period of one year, the material shall meet all requirements of this specification (see 4.6.8).

3.10 Marking for identification. Marking for identification shall be in accordance with MIL-STD-130 and shall include, but not be limited to, the following (see 4.6.1).

a. Title, number, and date of this specification.

b. Military part number.

c. Manufacturer's name or trademark.

d. Any necessary hazard warnings.

e. Mixing instructions.

f. Component identification (part A or part B).

g. Date of manufacture.

3.11 Workmanship. The material shall be free from lumps, foreign matter, or other visible defects that would detract from its appearance or function (see 4.6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection

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requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. Inspection and testing shall be classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed under ambient conditions.

4.4 First article inspection.

4.4.1 First article sample. When specified in the contract or purchase order (see 6.2), a first article sample of a quantity of adhesive sufficient to perform all tests specified herein shall be provided. The first article sample shall be fabricated using the same processes, facilities, and materials as are proposed for production. The first article shall meet all requirements of this specification.

4.4.2 Action in case of failure. No failures shall be permitted during first article sample inspection. If a failure occurs, the first article sample shall be rejected. No units shall be offered for acceptance until approval of a first article sample has been obtained from the procuring activity.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of all the tests listed in table II and as specified herein.

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TABLE II. Quality conformance inspection.

Examination and test	Requirement paragraph	Method paragraph
Material	3.2	4.6.1
Cure time	3.3	4.6.2
Pot life	3.4	4.6.3
Viscosity	3.5	4.6.4
Hardness	3.6	4.6.5
Shear strength	3.7	4.6.6
Flexural strength	3.8	4.6.7
Shelf life	3.9	4.6.8
Marking	3.10	4.6.1
Workmanship	3.11	4.6.1
Packaging	5	4.6.1

4.5.1 Lot size. Lot size shall consist of all the adhesive manufactured in one production batch run from one process forming part of one contract or order and submitted for acceptance at one time.

4.5.2 Sampling. Unless otherwise specified, sampling shall be performed on a quantity of adhesive sufficient to assure that all requirements of this specification have been met.

4.5.2.1 Action in case of failure. If a sample fails to pass all inspections, acceptance and shipment of the product shall be discontinued until corrective action acceptable to the procuring activity has been taken. After corrective action has been taken, inspections shall be repeated on an additional sample unit.

4.6 Test methods.

4.6.1 Visual inspection. Visual inspection shall verify compliance with 3.2, 3.9, 3.10, 3.11, and 5.

4.6.2 Cure time. Mix an aggregate of one ounce of adhesive and expose to a temperature of 65 ± 5°C (149 ± 9°F) for a period of four hours. Successful completion of the hardness test (specified in 4.6.5) shall assure compliance with 3.3.

4.6.3 Pot life. Mix a sample as specified in 4.6.2. Maintain at 25 ± 2°C (77 ± 4°F) for a period of one hour. If hardness of the material precludes spreadability, requirements of 3.4 have been met.

4.6.4 Viscosity. Viscosity shall be determined in accordance with ASTM D1084. Verify compliance with 3.5.

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4.6.5 Hardness. Shore D hardness shall be determined in accordance with ASTM D2240. Assure compliance with 3.6.

4.6.6 Shear strength. Shear strength shall be determined in accordance with ASTM D1002. Verify compliance with 3.7.

4.6.7 Flexural strength. Flexural strength shall be determined in accordance with ASTM D790. Verify compliance with 3.8.

4.6.8 Shelf life certification. Inspect the supplier's certification of shelf life capability and assure that requirements of 3.9 have been met.

5. PACKAGING

5.1 Packaging and packing. Packaging and packing shall be commercial type in accordance with ASTM D3951 (see 4.6.1).

5.2 Marking for shipment and storage. Marking for shipment and storage shall be in accordance with MIL-STD-129. In addition to the requirements of 3.10, marking shall include, but not be limited to, the following.

a. Lot or batch number.

b. Manufacturer's grade, class, or part number as applicable.

c. Storage requirements.

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use in various rocket motor applications.

6.2 Ordering data. Procurement documents should specify, but not be limited to, the following:

a. Title, number, and date of this specification.

b. The military part number.

c. Whether a first article sample is required (see 3.1, 4.4, and 6.3).

d. Inspection conditions if different from 4.3.

e. Sampling plan if at variance with 4.5.2.

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f. Quantity desired, by container size.

g. Place of delivery.

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of a quantity of adhesive sufficient to conduct all tests specified herein. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely upon such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (key word) listing.

Adhesive
Copolymer
Polyurethane

Custodian:
Army - MI

Preparing activity:
Army - MI

(Project No. 8040-A128)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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